two vertical lines. This arrangement is called a determinant.

In this chapter, we study determinants up to order three only with real entries. Also we shall study various properties of determinant (without proof), minors, cofactors, adjoint, inverse of a square matrix and business applications of determinants.

We have studied matrices in the previous class. Let us recall the basic concepts and operations on matrices.

1.1.1 Recall

Matrix

Definition 1.1

A matrix is a rectangular arrangement of numbers in horizontal lines (rows) and vertical lines (columns). Numbers are enclosed in square brackets or a open brackets or pair of double bars. It is denoted by A, B, C, …

For example , $A = \begin{bmatrix} 1 & 4 & 2 \\ 5 & 8 & 6 \end{bmatrix}$

Order of a matrix

If a matrix A has m rows and n columns, then A is called a matrix of order $m \times n$.

For example, If , $A = \begin{bmatrix} 1 & 4 & 2 \\ 5 & 8 & 6 \end{bmatrix}$ then order of A is 2×3

General form of a Matrix

Matrix of order $m \times n$ is represented as

$$\begin{bmatrix} a_{11}\ a_{12}\ a_{13} & \cdots\ \cdots\ a_{1i} & \cdots\ \cdots a_{1n} \\ a_{21}\ a_{22}\ a_{23} & \cdots\ \cdots a_{2i} & \cdots\ \cdots\ a_{2n} \\ \cdots\ \cdots\ \cdots & \cdots\ \cdots\ \cdots & \cdots\ \cdots\ \cdots \\ \cdots\ \cdots\ \cdots & \cdots\ \cdots\ \cdots & \cdots\ \cdots\ \cdots \\ a_{m1}\ a_{m2}\ a_{m3} & \cdots\ \cdots\ a_{mi} & \cdots\ \cdots\ a_{mn} \end{bmatrix}$$

It is shortly written as $[a_{ij}]_{mxn}$ $i = 1, 2, \ldots\ldots m$; $j = 1, 2, \ldots\ldots n$. Here a_{ij} is the element in the ith row and j^{th} column of the matrix.

Types of matrices

Row matrix

A matrix having only one row is called a row matrix.

For example

$$A = [\ a_{11}\ a_{12}\ a_{13} \ldots\ldots\ldots\ldots a_{1i} \ldots\ldots\ldots a_{1n}\ ]_{1 \times n};$$

$$B = \begin{bmatrix} 1 & 2 \end{bmatrix}_{1\ X\ 2}$$

Column matrix

A matrix having only one column is called a column matrix

For examples,

$$A = \begin{bmatrix} a_{11} \\ a_{21} \\ . \\ . \\ . \\ a_{m1} \end{bmatrix}_{m \times 1}, \quad B = \begin{bmatrix} 3 \\ -5 \end{bmatrix}_{2 \times 1}$$

Zero matrix (or) Null matrix

If all the elements of a matrix are zero, then it is called a zero matrix. It is represented by an English alphabet 'O'

For examples,

$$O = [0], \quad O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad O = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

are all zero matrices.

NOTE

Zero matrix can be of any order.

Square matrix

If the number of rows and number of columns of a matrix are equal then it is called a square matrix.

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